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The dead band in the performance of cross-flow turbines: Effects of Reynolds number and blade pitch

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ABSTRACT

The performance of a three straight bladed Cross-Flow or Vertical Axis Turbine based on symmetrical profiles, with a chord-to-diameter ratio of 0.16 has been studied in detail. The research is focused on the effects of Reynolds number and blade pitch angle, not only in the performance itself, but in the appearance or not of a dead band region in which aerodynamic loading cannot overcome resistive loads. Experiments were carefully designed in order to cover the transitional region were this dead band is incipient, for Reynolds numbers in the range from 3 to $5{\cdot}10^5$.

Results show how the performance is highly dependent on pitch angle at all Reynolds numbers investigated. Efficiencies over 25% are reached at tip speed ratios (blade to free stream velocity ratio) slightly larger than 2, for pitch angles in the range 4–8°. The appearance of the dead band is conditioned by the combination of Reynolds number and pitch angle.

1. Introduction

Cross-Flow Turbines (CFT), also known as vertical axis wind turbines or Darrieus rotors [1], are characterized by having their shaft perpendicular to the incoming flow and are designed for Tip Speed Ratios (TSRs) greater than one [2]. The TSR is a dimensionless number for the relationship between the tangential speed of the blade due to its rotation and the free stream velocity

$$\lambda = \frac{\omega D}{2u_\infty} \tag{1}$$

where ω is the rotational speed, D the turbine diameter, and u_∞ the free stream velocity. The interest on straight bladed CFTs has been growing over the last decades because they have several advantages over Axial-Flow Turbines (AFTs), namely: omni-directionality, lower maintenance costs, better behaviour in unsteady and skewed wind conditions with complex flow patterns such as high wind velocities and turbulent wind flows [3–5].

Performance curves result from plotting the power coefficient as a function of TSR, and are widely used to represent the efficiency of turbines. The power coefficient is the ratio of the mechanical power to the power available in the free stream, a dimensionless form of the power output which indicates the efficiency of the turbine, and is defined as

$$C_P = \frac{T\omega}{\frac{1}{2}\rho u_\infty^3 LD} \tag{2}$$

where T is the net torque applied on the shaft, ρ is the fluid density, and L the blade span. The shape of this performance curve in the case of cross-flow turbines shows three clearly distinct and well defined regions as a function of the tip speed ratio [6]. In the first region (R1), for the lowest TSRs, the power coefficient grows monotonically. For intermediate TSRs, in the second region (R2), the C_P exhibits its maximum values. In the third region (R3), for highest TSRs, the C_P drops abruptly. For sufficiently low or high λ, the performance of this type of turbines is considerably lower than that typical of region R2. Besides, it is also well known that the higher the turbine solidity $\left(\sigma = \frac{Nc}{\pi D}\right)$, which depends not only in the diameter but also in the number of blades (N) and the chord (c), the lower the optimal tip speed ratio that maximizes its performance [7–9]. Important parameters with great influence on performance of turbines are the pitch angle between the foil centreline and the line tangent to the circular trajectory of the blade at its mid-chord (β) and the Reynolds number based on the turbine diameter ($Re_D = u_\infty D/\nu$, with ν being the kinematic viscosity of the fluid).

In order to obtain the performance curve of a three straight bladed CFT, Bachant and Wosnik [10] towed their model in a water tank. The turbine had blades with NACA-0020 profile and a chord-to-diameter ratio (c/D) of 0.14. Their experiments were all conducted at the same speed, with a constant Re_D of $1.0{\cdot}10^6$. Tip speed ratio was prescribed and

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held constant during each run, by using a servo motor coupled to the turbine shaft. They showed a maximum power coefficient of 0.26 at a tip speed ratio of 1.9. By changing Re_D in the range from $3.0{\cdot}10^5$ to $1.3{\cdot}10^6$ [11], they observed a drastic improvement in the dimensionless power output of the CFT.

Bravo et al. [12] used a three straight bladed CFT model with NACA-0015 blades and a c/D of 0.16 to carry out experiments in a full-scale wind tunnel for a range of diameter Reynolds number from $1.0{\cdot}10^6$ to $2.7{\cdot}10^6$. They measured the torque with a load cell and devised an active closed-loop speed control system, that accurately regulated the rotational velocity of the rotor. They controlled the speed because the system was not self-regulating at the lowest TSR of the performance curve, before reaching the maximum power output. With a constant load the CFT was not able to keep its angular rotating speed ω, so it tended to slow down until stopping or headed to higher speeds. At all Re_D investigated, the maximum power coefficient occurred at a tip speed ratio $\lambda \approx 1.6$ and for $Re_D \geqslant 1.3{\cdot}10^6$ the performance curve became essentially Re independent. Remarkably, they did not obtain their maximum C_P, with a value of 0.3, at the highest Reynolds number, but at $Re_D = 1.7{\cdot}10^6$. Fiedler and Tullis [13] conducted identical experiments but at lower Reynolds numbers, $8.3{\cdot}10^5 < Re_D < 1.8{\cdot}10^6$, showing consistent results. They repeated the measurements with different preset pitch angles at $Re_D = 1.7{\cdot}10^6$, taking into account that a forward shift of mount-point from the centre of the chord towards the leading edge corresponds to an effective toe-in pitch change of

$$\beta = tan^{-1}\left(\frac{2x}{D}\right) \tag{3}$$

where x is the distance between these two points along the blade chord. With a $\lambda = 1.6$, the authors obtained a better performance as they changed β from 6.4° toe-in to 5.3° toe-out, passing through 2.5° toe-in and 1.4° toe-out.

In the work by Takao et al. [14], the authors used a three straight bladed CFT, but this time based on asymmetric NACA-4518 profiles, with a c/D ratio of 0.17. The experiments were performed in an open-jet wind facility at $Re_D = 3.2{\cdot}10^5$, and a maximum power coefficient $C_P = 0.11$ was obtained at a tip speed ratio $\lambda = 1.7$. Li et al. [15] investigated a four straight bladed CFT with a chord-to-diameter ratio $c/D = 0.13$ and NACA-0021 blades. With a pitch angle of 8° toe-out, the turbine had the highest power output. In addition to a torque meter, the authors used a six-component balance to measure the loads on the turbine at a $Re_D = 1.1{\cdot}10^6$. The power coefficient slowly rose up to $C_P \approx 0.15$ and then from $\lambda = 1.7$, it decreased rapidly. Li et al. [16] repeated the previous experiment on a two straight bladed CFT model with a c/D ratio of 0.13, changing the pitch angle in 2° steps from $\beta = 4°$ toe-out to $\beta = 8°$ toe-out. The turbine performance increased up to $C_P = 0.21$ at $\lambda = 2.2$ when adjusting the blade pitch from 4° to 6° toe-out, but decreased when moving β from this last value to 8° toe-out. Moreover the optimal tip speed ratio that maximized its performance was slightly increased as the pitch angle became higher.

El-Samanoudy et al. [17] with their four straight bladed CFT based on NACA-0024 airfoil with a chord-to-diameter ratio of 0.19 in a wind open-jet facility at a $Re_D = 4.3{\cdot}10^5$, also observed that the maximum dimensionless power output began to decrease as soon as the pitch was configured with toe-out positions. Using load cells to measure the tension of a rope in contact with a grooved pulley attached to the shaft, they found a maximum power coefficient of 0.25 at $\lambda = 1.3$ for a β of 10° toe-out. The maximum C_P for both $\beta = 0°$ and $\beta = 20°$ toe-out, was smaller and took place at lower TSR. Strom et al. [18] tested straight bladed CFTs with two and four NACA-0018 blades with a c/D ratio of 0.24 in a water flume facility at $Re_D = 1.2{\cdot}10^5$, covering the range of pitch angles $0° < \beta < 12°$ toe-out with increments of 2°. They operated a servo motor by setting up constant velocities and measured the turbine torques with a six-axis load cell. In both cases the best scenario in terms of performance took place with $\beta = 6°$, obtaining the maximum C_P with a value of 0.18. The optimal tip speed ratio ($\lambda = 1.8$) for the two-bladed CFT was higher, but in both cases the performance curves had two clearly defined peaks. The region between the two peaks, where the torque can be negative was named *dead band* by Baker [19].

Araya and Dabiri [20] investigated a three straight bladed CFT with its blades based on NACA-0018 profile and a chord-to-diameter ratio $c/D = 0.33$, in a water channel facility. Their goal was to prove that motor-driven and flow-driven turbines have similar hydrodynamics. They defined the rotor *neutral curve* as the $\lambda(Re_D)$ curve that shows the upper bound of tip speed ratios for flow-driven cases, i.e. the maximum λ values that the rotor reaches if flow-driven. Their empirical curve was shifted to lower TSRs than that would be theoretically possible in the presence of only airfoil lift and drag forces because of the inherent losses of the turbine system, and was monotonically increasing except from around $Re_D \approx 6{\cdot}10^4$ where there was an abrupt rise. Moreover they obtained good agreement between the torque measurements of motor-driven and flow-driven cases using a rotary torque sensor at a diameter Reynolds number $Re_D = 7.8{\cdot}10^4$, when operating on the back side of the performance curve (for λ larger than those at the maximum power output) where CFTs operate with constant angular velocities. Edwards et al. [21] tested a three straight bladed CFT (NACA-0022 with a ratio $c/D = 0.06$) in a wind tunnel with an open section at a $Re_D = 3.1{\cdot}10^5$. They used a motor that could be engaged/disengaged via an electro-magnetic clutch, to drive the turbine up to high speeds passing through the dead band.

The objective of the work presented here, is to describe the effect of Reynolds number and blade pitch on the efficiency of cross-flow turbines. Although this has been investigated in the past [13,16–18], there are no published works that cover the transitional region in which the aerodynamic torque of the blades overtakes the resistive torques, with cases clearly showing the appearance of the dead band. We demonstrate how small variations of the pitch angle can change the shape of the performance curves, and therefore the behaviour of this type of turbines. Moreover we put in perspective these results, with recent measurements that show the flow dynamics inside the rotor of this type of machines.

2. Experimental set-up

The experiments were carried out in the Boundary Layer Wind Tunnel (BLWT) of the Laboratory for Fluid-Structure Interaction (LIFE), at Universitat Rovira i Virgili (URV) in Tarragona. The facility is an open-circuit blower tunnel, with an axial fan that forces the air into a settling chamber. The settling chamber is composed of an array of three honeycombs that reduce turbulence intensity, and supply a more uniform flow to a three-dimensional contraction before the working section. Wind speeds up to 12 m/s can be reached. The wind tunnel has a total length of 23 m and a working section of $1.84 \times 1.22\ m^2$ with 14 m of optical access. The model was placed centred in the working section in order to avoid wall effects, and to have a uniform flow profile. The wind tunnel calibration in the test-section previous to the experiments presented here, resulted in a turbulence intensity of 2.4% in the worst case scenario, i.e. at the highest Reynolds number in the experiments.

The turbine model consisted of a three straight bladed cross-flow turbine with its blades based on the NACA-0015 profile, printed in PolyLactic Acid (PLA). The blades had a chord of 0.12 m and a total span of 0.75 m with an inner core made of an aluminium 8 mm square tube, that provided very high flexural stiffness to the blades. They were connected to a horizontal stainless steel shaft with a diameter of 25 mm, through six aluminium struts located at 1/5 and 4/5 of the blade span. The turbine diameter (D) was 0.75 m leading to a chord-to-diameter ratio (c/D) of 0.16 and a span-to-diameter ratio (L/D) of 1. The parts connecting the aluminium square tube that acted as the core of the blades and the two struts, were also printed in PLA and allowed the blades to be oriented with the desired fixed pitch angle. Experiments were conducted by setting up 9 different pitch angles β, taking into account the effective toe-in pitch in Eq. (3) for a forward shift of the

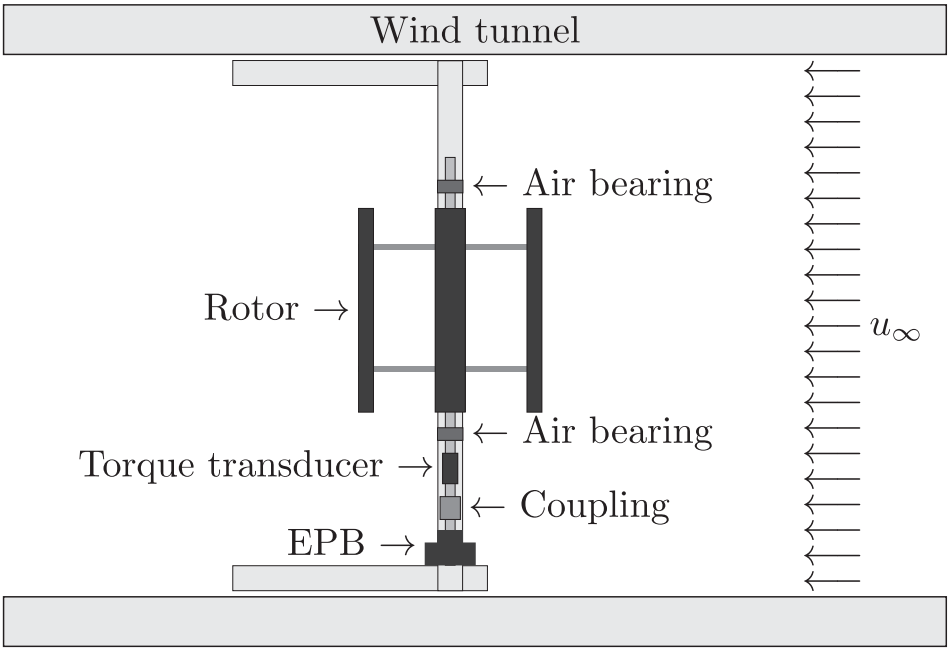


Fig. 1. Layout of the experimental set-up in the wind tunnel test section.

mount-point. The angles were 8° toe-in, from 0° to 12° toe-out in steps of 2°, and finally 16° toe-out. For each β, tests were conducted at 3 different turbine diameter Reynolds numbers of 3, 4 and $5{\cdot}10^5$, yielding a total of 27 experimental cases.

The rotor shaft was horizontally supported by two air bearings that ensured an extremely low friction when rotating, so dry friction losses were negligible in these experiments. A rotating torque transducer with a measurement range of 5 N m with an accuracy of 0.20% of its full scale range, was installed at one of the shaft ends. At the other side of the sensor, an Electromagnetic Powder Brake (EPB) with a capacity of 2 N m, was installed. The brake allowed to apply torques to the turbine shaft, with an accuracy of 0.005 N m. The only rotational frictions acting on the rotor during its operation were due to the low friction bearings included in the EPB. Fig. 1 shows a schematic top view of the experimental layout in the wind tunnel test section.

The free stream velocity (u_∞) in the wind tunnel was measured during each experiment by means of a Pitot tube. A thermocouple installed at the same point, allowed the computation of a corrected air density and kinematic viscosity. The angular velocity of the rotor ω, was measured with an analogue encoder geared to the rotor shaft. In order to know the azimuthal position of the turbine, a magnetic proximity sensor installed in one of the blade struts was also used. For the experiments all signals were sampled at a rate of 2 kHz during at least 60 s.

3. Results and discussion

In this section we first describe the method leading to the obtention of the performance curves. The test example in Fig. 2, is used for the purpose. The dynamics of the rotor can be described by the moment equation

$$T_B + T_R + T_A = I\alpha \tag{4}$$

where T_B is the blade aerodynamic torque generated by the action of the wind on the blades being the only positive value of the expression. T_R accounts for the resistive torques in the shaft (including all dry friction elements in the EPB and the aerodynamic drag of the struts that support the blades), and T_A is the external applied torque (or load torque) resulting from the operation of the electromagnetic powder brake. In the right hand side of the equation there is the time variation of angular momentum in the shaft, expressed as the total shaft inertia (I), assumed to be constant, times the angular acceleration ($\alpha = \frac{d\omega}{dt}$). The right hand side of the equation is zero when the angular velocity of the shaft is constant, and this happens when all the load torques (resistive + brake applied load) balance the aerodynamic torque produced by the blades. On the other hand, if ω is not constant, the rotor is either accelerating or decelerating. It will decelerate if $|T_R + T_A| > T_B$, meaning that the aerodynamic torque is smaller than the existing loads acting on the rotor. The machine will only self-start accelerating if $|T_R| < T_B$, assuming that no braking torque is applied during this situation. In this manuscript, we will show performance curves obtained after the measurement of the aerodynamic blade torque (T_B), therefore representing the net torque acting on the rotor.

Plot a) of Fig. 2 shows the sum of the brake torque applied to the system plus the torque resulting from the variation of the angular momentum in the shaft, as a function of the angular speed of the rotor ($T_B + T_R = I\alpha - T_A$). We use in the rest of the manuscript the terms front side and back side of the curve, in order to refer to the part of the curve to the left or to the right of the maximum torque, respectively. On the back side of the curve, the rotor is characterised by having a constant ω that depends on the external brake load applied, therefore Eq. (4) reduces to ($T_B + T_R = -T_A$). On the front side of the curve, with ω smaller than that at the maximum torque, the rotor cannot reach constant angular velocity states and decelerates. The value of the deceleration depends on the applied external braking load T_A.

The experiments were carried out by configuring the desired free stream (or Re_D) in the wind tunnel. A DC motor geared to the rotor shaft through a 1: 1 transmission, was used to drive the turbine up to an ω of approximately 70 rad/s, moment at which the DC motor was disengaged from the rotor shaft. During this process the brake was obviously not acting on the system ($T_A = 0$). The turbine then started to decelerate down to a steady condition given by a constant rotational speed. This spontaneous stabilization of the turbine was reached when the aerodynamic torque acting on the blades balanced the very small opposing frictional torque (mechanical parts rotating in the non-operating EPB) plus the resistive aerodynamic torque (drag on the rotating struts) acting on the system. In this state, without external loads applied to the rotor and spinning at its maximum ω, the time-averaged torque was zero. From that point on, the back side of the curve is obtained by applying increasing brake torques (T_A). In the experiments this was

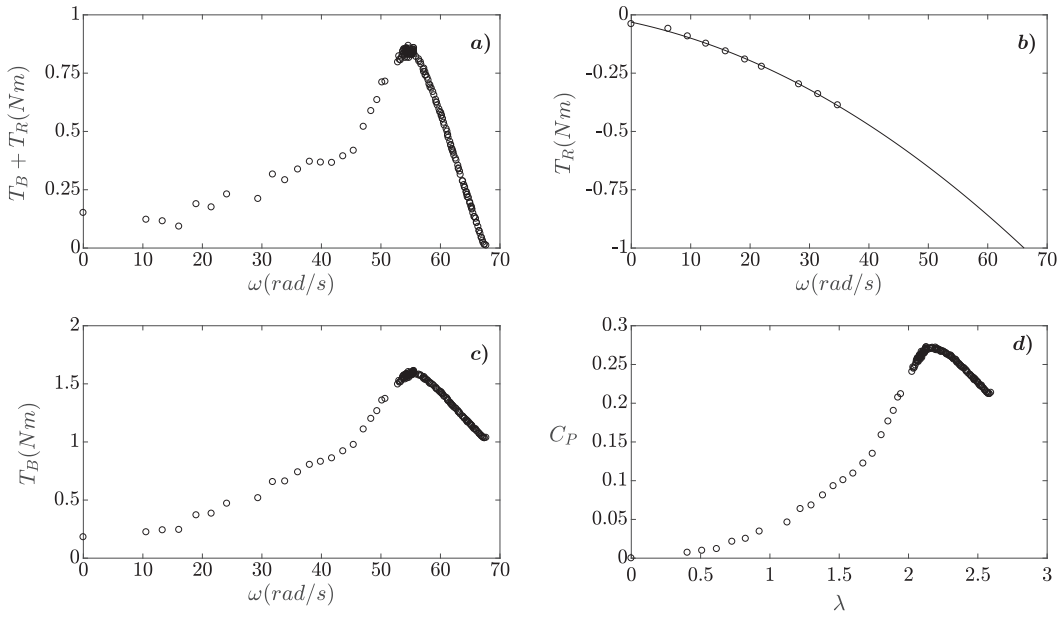


Fig. 2. Example of torque and performance curves obtained during this experimental campaign. From left to right and from top to bottom: $T_B + T_R(\omega)$ curve, $T_R(\omega)$ curve, $T_B(\omega)$ curve, and performance curve.

done by changing the load applied by the EPB in steps of approximately 0.005 N m and leaving time enough for the rotor to reach a new constant ω, when the blade aerodynamic torque balanced the brake torque. After each of these brake torque changes, the torque was measured during at least 60 s in order to be able to compute accurate time-averaged values. Eventually, a maximum value of torque was reached that if surpassed led to the start of the front side of the curve, with the rotor slowly decelerating to rest. Once the maximum torque was surpassed, the velocity started to decrease, i.e. the aerodynamic torque was not large enough to balance the applied brake torque. In this part, the curve has a positive slope and Eq. (4) applies. Recall that $I\alpha$ value is 0 on the back side, but negative on the front side of the curve. In order to obtain the points in this part of the curve, the time series of the signals recorded, were processed using time windows of 1 s, in a quasi-steady approach. This results in a moving averaged ω, that in fact indicates how the deceleration between two of these consecutive time windows takes place. The same was done with the torque signal. This methodology has been used by Edwards et al. [21] to obtain performance curves.

In order to obtain the net aerodynamic blade torque (T_B), the resistive torque (T_R) was needed beforehand. We carried out a series of tests at each Reynolds number, but this time without the blades installed in the rotor. The system without blades was driven with the DC motor at different ω while measuring the torque on the shaft. During this type of tests, the torque load cell measured only the aerodynamic drag due to the rotating struts. By measuring at different rotational speeds, the resistive torque curve as a function of ω for each Re_D was obtained. A typical curve appears in plot (b) of Fig. 2. The values of the resistive torque are negative, with a quadratic trend imposed by the aerodynamic drag on the struts, as in the experiments presented here dry friction was minimal because the shaft was supported by air bearings. By removing this resistive torque curve from the measured torque during the experiments, depicted in Fig. 2(a), we obtained the net blade aerodynamic torque (T_B) that is shown in plot (c) of the same figure. Note that by streamlining the struts in order to reduce their aerodynamic drag, more points could have been obtained in the stable back side of the curve at higher ω. However it was not the intention of this work to produce an optimised machine in terms of design, but to understand the performance variations associated to the pitch angle variation and its relationship with the dead band in case of appearance.

Plot (d) of Fig. 2 shows the final performance curve of the turbine for the example presented, i.e. the dimensionless power coefficient calculated using T_B in Eq. (2) as a function of the tip speed ratio (Eq. (1)).

3.1. Effect of Reynolds number

Having described how the results were obtained, in this section we show how performance curves vary with Re_D. We have analysed one single pitch angle for the purpose: $\beta = 8°$ toe-out. The left side plot in Fig. 3 shows the performance curves for the selected blade pitch at three different Re_D. At the lowest diameter Reynolds number ($Re_D = 3{\cdot}10^3$), the CFT accelerated self-starting from rest up to a $\lambda = 1.5$, reaching a maximum power coefficient $C_{Pmax} \approx 0.06$. Even if the cross-flow turbine was motor-driven to $\lambda > 1.5$, once the motor was disengaged, the turbine decelerated returning to a steady λ of 1.5. The aerodynamic torque generated by the blades could only overcome the resistive torque at very low TSRs.

With a Re_D of $4{\cdot}10^5$ the rotor, if not driven by the motor, accelerated from rest up to $\lambda \approx 1.7$. If the motor was used to drive the turbine shaft to high tip speed ratios and disengaged afterwards, the CFT stabilized at $\lambda = 2.6$ producing a C_{Pmax} of 0.23 at λ around 2.4. The change in slope (or plateau from now on) that appears in the figure for the range of TSRs $1.7 < \lambda < 1.9$ shows the so-called *dead band* where the useful blade torque is lower than the resistive torque. In a torque curve, this dead band region would appear with negative values. The fact that the curve for $Re_D = 4{\cdot}10^5$ evolves parallel to the one with the highest Re_D, is a consequence of the dead band and the C_P plateau just described.

At the highest diameter Reynolds number $Re_D = 5{\cdot}10^5$, the dead band totally disappeared and the cross-flow turbine accelerated self-starting from rest up to a $\lambda = 2.6$, reaching a maximum power output over 0.27 at a λ near 2.1. This self-starting behaviour implied that the blade aerodynamic torque was higher than the resistive torque at all times. For all Re_D, the front sides of the performance curves where the CFT is able to accelerate without any external torque other than the aerodynamic one, can be approximated by a cubic relationship of expression $C_P = a\lambda^3$, with the coefficient a increasing with Re_D.

The plot on the right column of Fig. 3 shows the neutral curve defined on Section 1, for the case with blade pitch $\beta = 8°$ toe-out. The curve shows the maximum λ values that the rotor is able to reach if flow-driven and without brake torque applied to it, starting from rest.

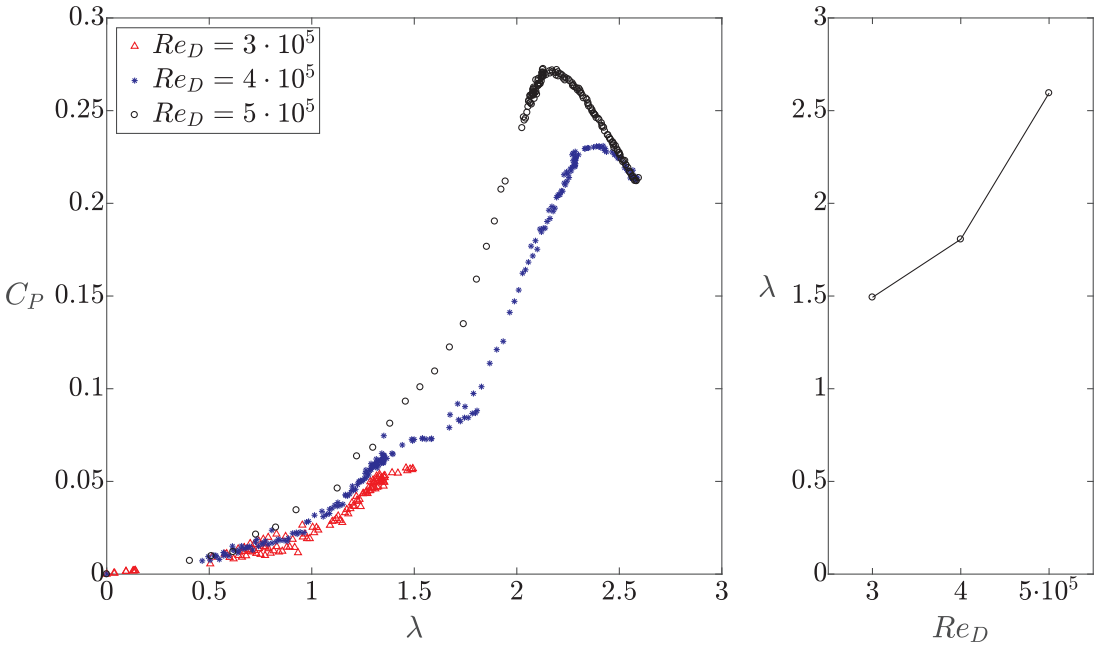


Fig. 3. From left to right: performance curves at different diameter Reynolds numbers, and turbine neutral curve for $\beta = 8°$ toe-out.

The change of slope from $Re_D = 4{\cdot}10^5$ to $5{\cdot}10^5$ can be explained because the aerodynamic torque produced by the blades surpasses the resistive torque for all the range $0 < \lambda < 2.6$ without the need of forced-starting.

The behaviour of the cross-flow turbine with increasing Re_D described here for $\beta = 8°$ toe-out can be extrapolated to all the blade pitches tested. As the Reynolds number was increased from $3{\cdot}10^5$ to $5{\cdot}10^5$, the torque curve would show positive values in a larger range of TSRs. At low Re_D, T_B is higher than T_R only at low λ; whilst at mid Re_D, there are two different ranges of tip speed ratios with positive torque values separated by a dead band that can be identified in the performance curves as a C_P plateau. Finally, at the highest Re_D tested, the dead band disappears completely, allowing high power coefficients over a larger range of TSRs and moving the maximum C_P to a lower λ. The Re_D that limits these three different behaviours will depend on the blade pitch angle, as it will be detailed in the next section.

3.2. Effect of blade pitch

In order to study the effect of blade pitch on the performance and on the existence or not of the dead band, we compare here the curves for nine different blade pitches in the range from $\beta = 8°$ toe-in to 16° toe-out at three different constant turbine diameter Reynolds number. Fig. 4 shows the case for the Re_D of $3{\cdot}10^5$. The results depict large changes in performance with very small variations of the pitch angle. The lowest performance took place for $\beta > 8°$ toe-out, with the turbine not being able to reach TSR larger than 0.5. The same happened at the other end of the range, for $\beta = 8°$ toe-in. With a $\beta = 0°$, the power coefficient increased considerably, although when the DC motor was disengaged at high TSRs, the turbine decelerated down to a stable $\lambda \approx 1.5$. A very similar behaviour was found for $\beta = 8°$ toe-out. Remarkably for pitch values in the range from 2° to 6° toe-out, when the DC motor was disconnected from the rotor shaft, the CFT stabilized at a TSR near 2.5, reaching a C_{Pmax} of 0.25 at a λ near 2.3 for $\beta = 4°$. The change in the curve slope (or the plateau) in the C_P near a TSR of 1.5, indicates the existence of a dead band for all three cases. The fact that the curve for the blade pitch of 6° is displaced to the right, climbing parallel to the 2° and 4° curves, after a wider plateau, means that the dead band was more significant when the pitch was 6°.

The data for a Reynolds number of $4{\cdot}10^5$ appears in Fig. 5. The scenario is very similar to that reported for the smallest Re_D, with the lowest performance cases in the range from 8° toe-in to 0° and from 10° to 16° toe-out. At this slightly higher Re_D, the blade pitch of 8° toe-out is also producing large efficiencies. For $\beta = 4°$ and $\beta = 6°$ toe-out, the CFT was able to self-start from rest up to $\lambda = 2.5$ without the appearance of a dead band, producing a maximum value of C_P near 0.25 at a λ around 2.2 for $\beta = 6°$. The curve appears displaced to the right for $\beta = 2°$ toe-out, and parallel to these other three curves, with a wider dead band, there is the curve for blade pitch 8° toe-out.

Finally, the results for the highest Reynolds number investigated, $Re_D = 5{\cdot}10^5$, are presented in Fig. 6. As in the cases of Fig. 5, four blade pitches ($2° < \beta < 8°$ toe-out) result in curves with power coefficient maxima over 0.1. For $\beta = 8°$ toe-out, the rotor accelerated self-starting from rest up to $\lambda = 2.6$, reaching a C_{Pmax} over 0.27 at a λ near 2.1. As the Re_D is increased, the optimal TSR at which the maximum power coefficient takes place is lower. In this case, the cubic relationship $C_P = a\lambda^3$ that characterises the front side of the performance curve, has a larger a coefficient. For β in the range from 2° to 6° toe-out, the maximum power coefficient decreased, whilst in the range 8° to 10° this value was higher than at the lower Re_D investigated. This means that, apart from improving the performance of the cross-flow turbines, as we increased the velocity of the incoming flow, the optimal fixed blade pitch was higher.

Fig. 7 shows the maximum dimensionless power output as a function of the pitch angle for the three Re_D studied. The vertical black line indicates $\beta = 0°$ for reference. For $Re_D = 3{\cdot}10^5$, there is a sharp peak with a local maximum for angles between 2° and 6°, and C_P values over 0.2 and up to 0.25. As the turbine diameter Reynolds number is increased up to $Re_D = 4{\cdot}10^5$, the maxima of the C_P shows a wider region of high performance with values near 0.25 that extends from $4° < \beta < 8°$ toe-out. At the highest Re_D of $5{\cdot}10^5$, this peak has clearly moved to higher blade pitch angles with its maximum C_P over 0.27 for $\beta = 8°$ toe-out.

Several authors have used servo motors to obtain performance curves [10–13,18], but to the knowledge of the authors, there are no publications that discuss in-depth as in here, the changes in performance as a function of Reynolds number and pitch angle together with the appearance or not of the dead band. Most of the published works examine the performance of specific machines, but have not given an in-depth analysis of the phenomena shown here, in the transitional region of Re_D, in which the dead band appears or not depending on the

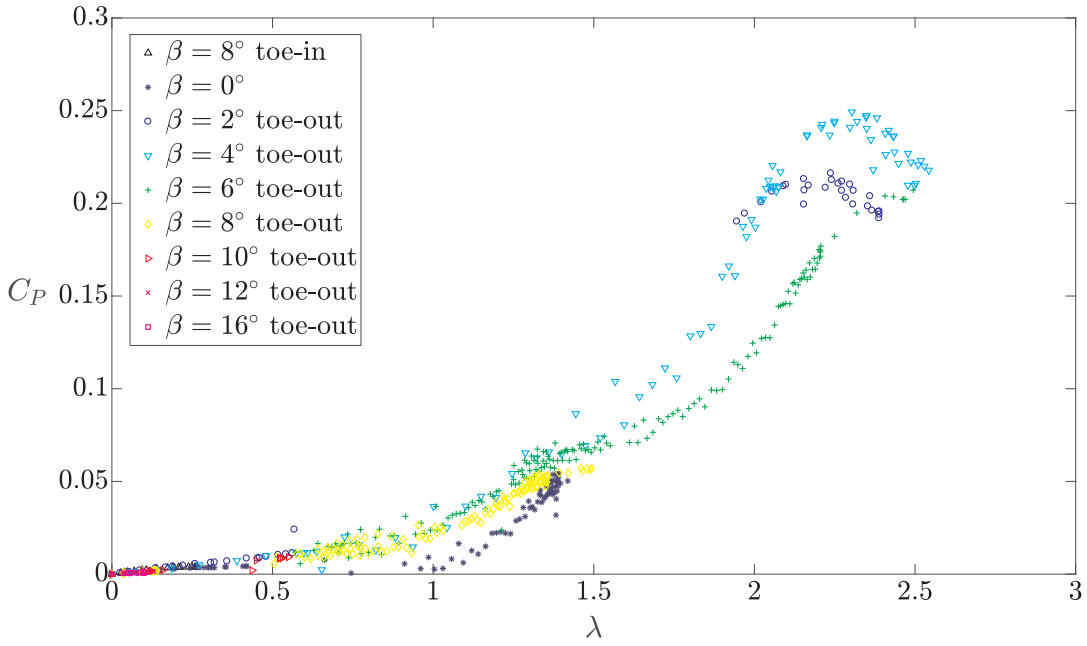


Fig. 4. Performance curves for different pitch angles at $Re_D \approx 3{\cdot}10^5$.

pitch angle. We demonstrate the implications the pitch angle has at different Re_D at which the dead band is incipient. This obviously has tremendous consequences not only for the overall performance of these machines, but also for their ability to self-start and to sustain their operation, when flow resources are scarce. The novelty of the present work lies in the fact that performance curves have been obtained in the dead band transitional region, without forcing the rotation of the turbine, and allowing the detailed study of the effects of *Re* and β on the performance of the machine. The presence of the dead band where the resistive torque is larger than the blade torque at lower or higher Re_D, depends clearly on the blade pitch angle imposed.

On top of that, in recent studies by Somoano and Huera-Huarte [6], and Somoano and Huera-Huarte [22], the authors conducted DPIV measurements inside the rotor of a similar CFT. They showed how the blade pitch angle had an enormous influence on the effective angle of attack seen by the turbine blade and hence, on the formation of the Leading-Edge Vortex (LEV) that is shed from the blades as they pass through the upstream part of the turbine. The timing of the LEV shedding was found to be crucial for how the blade-wake interactions took place in the downstroke region and for the dynamic stall characterization of the system. The study showed how for β near 8° toe-out, the LEV travelled closer to the blade suggesting larger forces on it, and therefore a constructive flow interaction. The results presented here, show that independently of Re_D, there is a β range that implies high performance ($C_P > 0.2$) if TSRs are near or over 2. Somoano and Huera-Huarte [6], and Somoano and Huera-Huarte [22], showed how at these λ, the flow around the blade in the downstroke region causes angles of attack high enough to suggest the production of large lift forces avoiding large drag forces, therefore resulting in high torques. For lower TSRs, the authors showed how the wake is dominated by the free

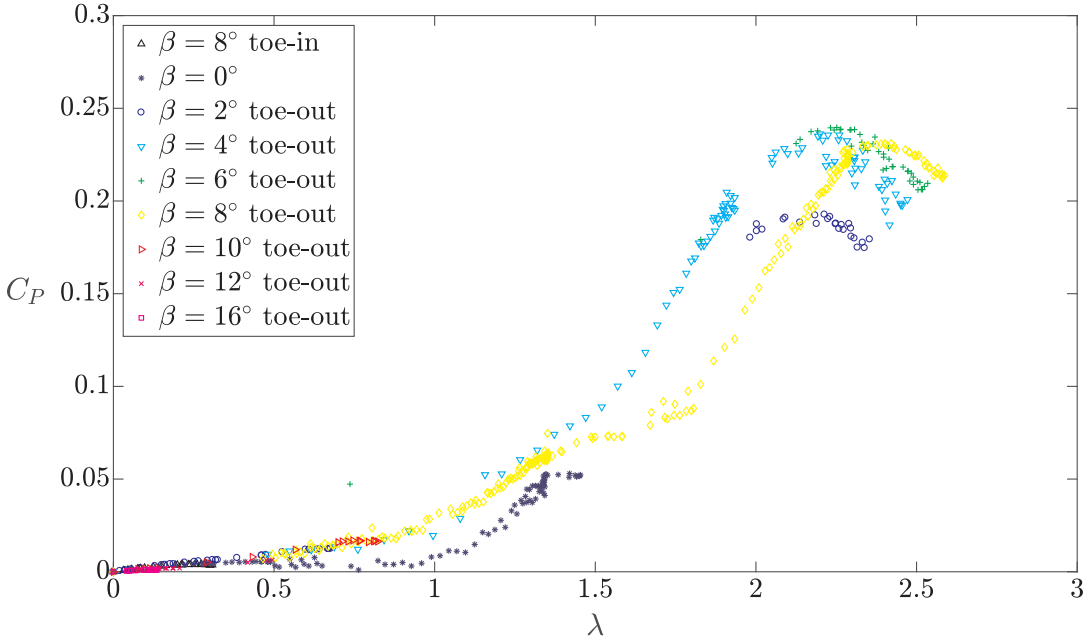


Fig. 5. Performance curves for different pitch angles at $Re_D \approx 4{\cdot}10^5$.

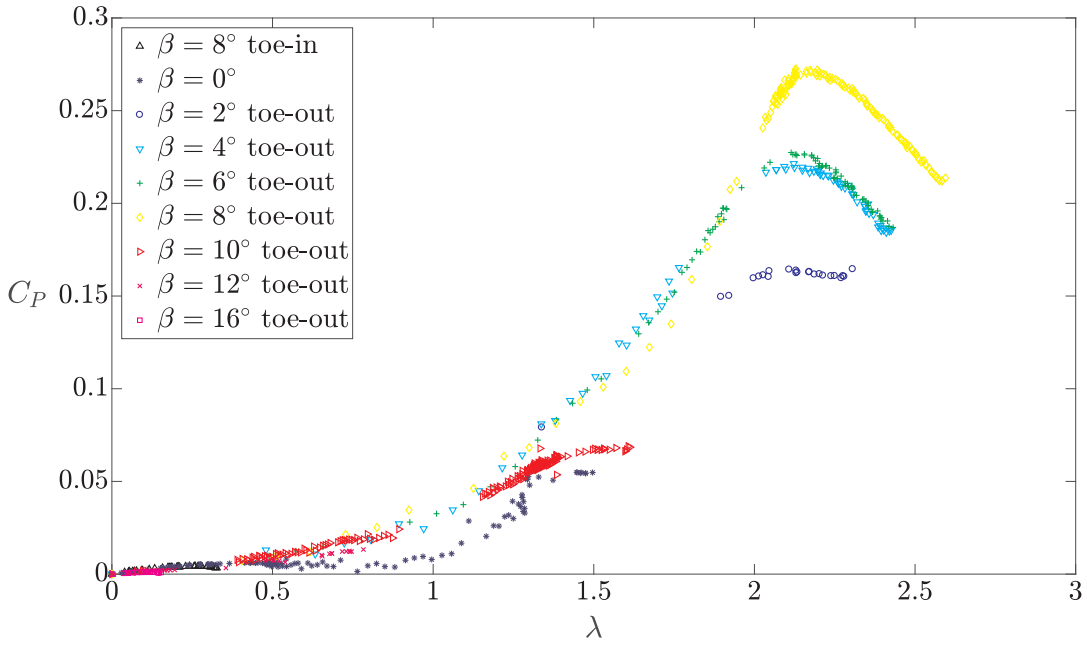


Fig. 6. Performance curves for different pitch angles at $Re_D \approx 5{\cdot}10^5$.

stream, promoting a local flow with a significant radial component, and hence yielding high drag forces in the downstroke part of the rotor. At the highest λ, the wake starts being dominated by the rotational motion of the machine implying low lift forces.

4. Conclusions

Experiments have been conducted in a wind tunnel with a cross-flow turbine model based on symmetric profiles in order to study the combined effects of Reynolds number and blade pitch, not only in the performance curves, but also on the existence or not of a dead band region. Experiments were carried out with Reynolds number based on diameter, in the range from $3{\cdot}10^5$ to $5{\cdot}10^5$, where the performance dead band is highly dependent on blade pitch. Blade pitch was varied in a wide range, from 8 degrees toe-in up to 16 degrees toe-out, with smaller increments from 0 to 12 degrees.

The turbine model showed the highest performances, when blade pitch was configured with angles from 4 to 8 degrees toe-out. These values are near 25% independently of Reynolds number. As the Reynolds number is increased the pitch angle associated to the highest performance becomes larger, and the region in which high performances are found cover a wider range of angles.

On the other hand, results show how at the smallest Reynolds number investigated in this transitional region, the dead band is more prone to appear. The appearance of the dead band can be recognised in the performance plots as curves are offset in parallel to larger tip speed ratios, after a clear change in the performance curves or plateau. In the absence of dead band the power coefficient grows following an expression with the form $C_P = a\lambda^3$, with the coefficient a being larger as the Reynolds number is increased, yielding optimal performances with

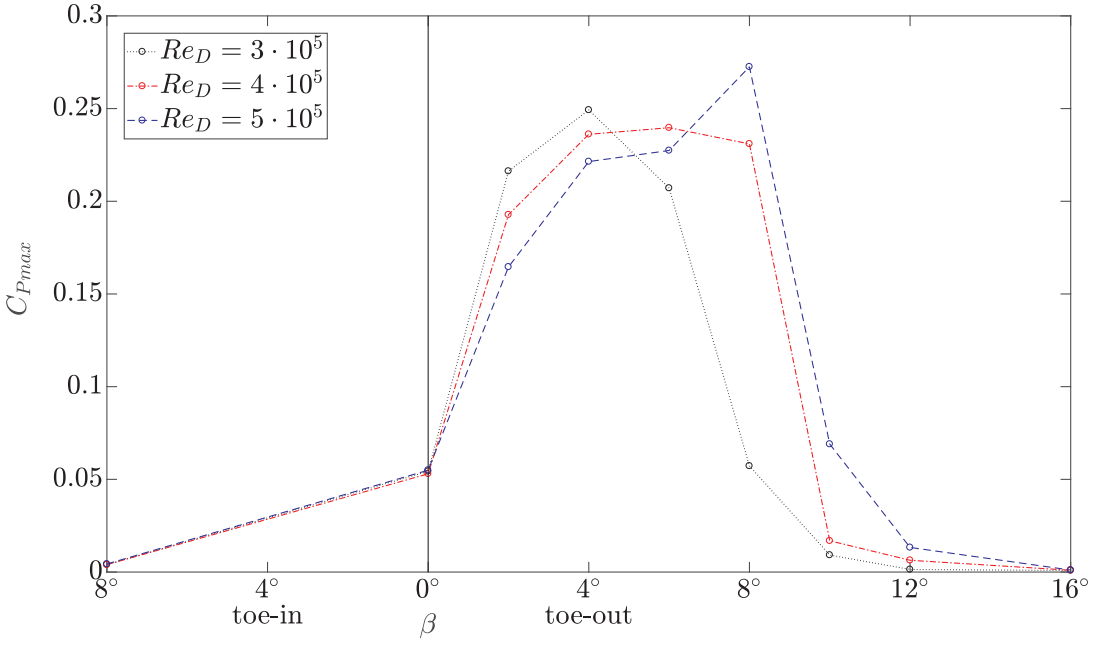


Fig. 7. Maximum power coefficient obtained as a function of the blade pitch for different turbine diameter Reynolds numbers.

smaller tip speed ratios.

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